

Cost-Aware Conditional Diffusion Compression of 4K-5K Drone Imagery

Jooho Kim, Anish Shakya, Yifan Yang, and Jacob Kelly

Texas A&M University

Problem

Post-disaster drone flights produce high-resolution imagery that is useful for inspection, mapping, and damage assessment. The cost appears downstream: full-resolution diffusion decoding moves slowly, uses large GPU memory, and makes repeated HPC workflows expensive. We ask a systems question: which operating point keeps native visual detail while reducing runtime and memory?

Decision Equations

$$R = \frac{24}{\text{BPP}}$$
$$S_T = 1 - \frac{T_{\text{tile}}}{T_{\text{full}}}, \quad G_M = \frac{M_{\text{full}}}{M_{\text{tile}}}$$
$$C(\theta) = \alpha \hat{T} + \beta \hat{M} + \gamma(1 - \text{SSIM})$$

The score is a transparent decision aid, not a universal model metric.

Key Result

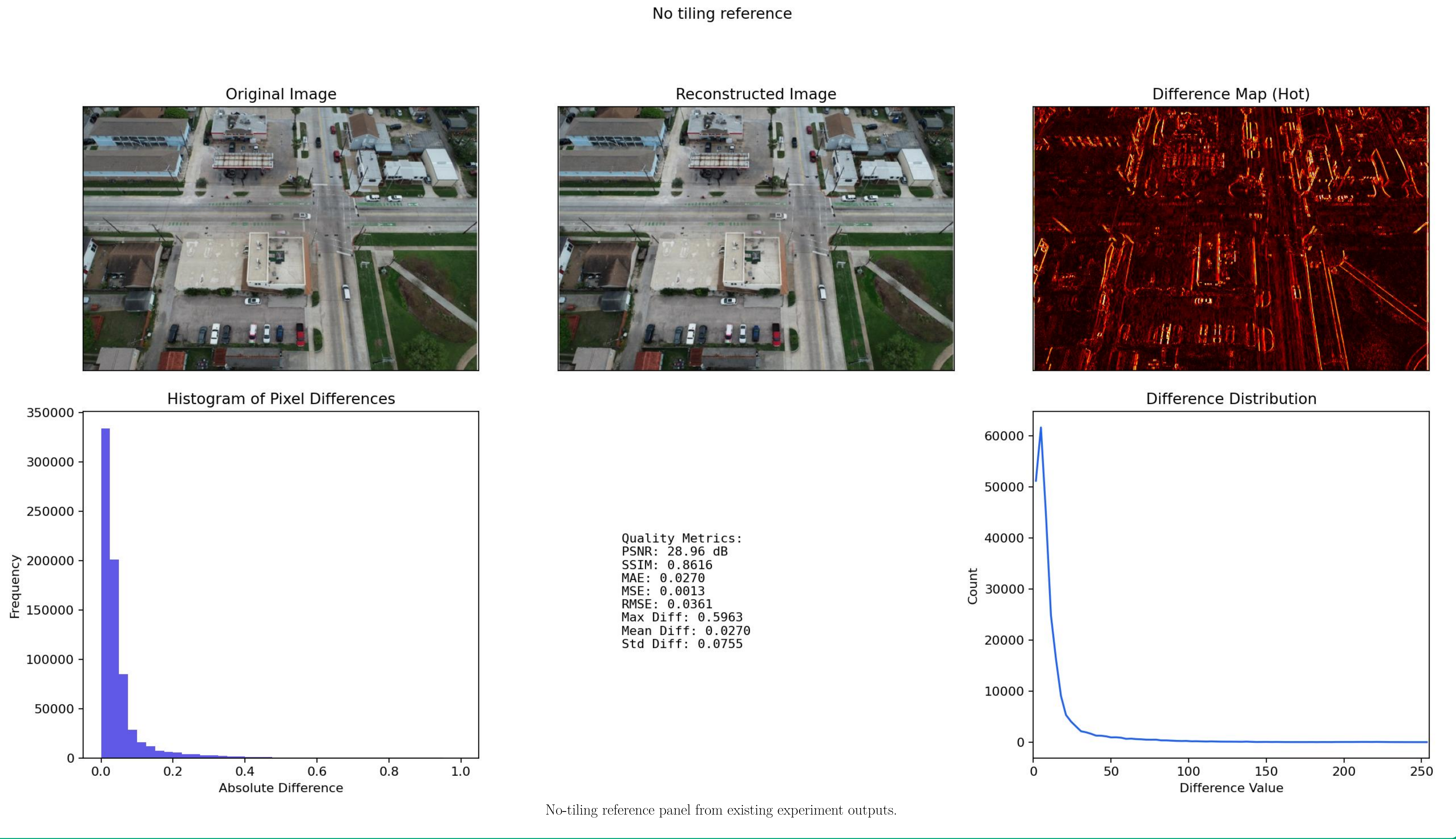
44.7%
runtime reduction

31.3×
lower peak memory

0.0031
SSIM drop vs 512 tile

The selected 256×256 tile is the current speed and memory candidate. It preserves the native image boundary and lowers peak memory from 52.01 GB to 1.66 GB on 50 full-resolution images.

Existing Tiling QA Evidence



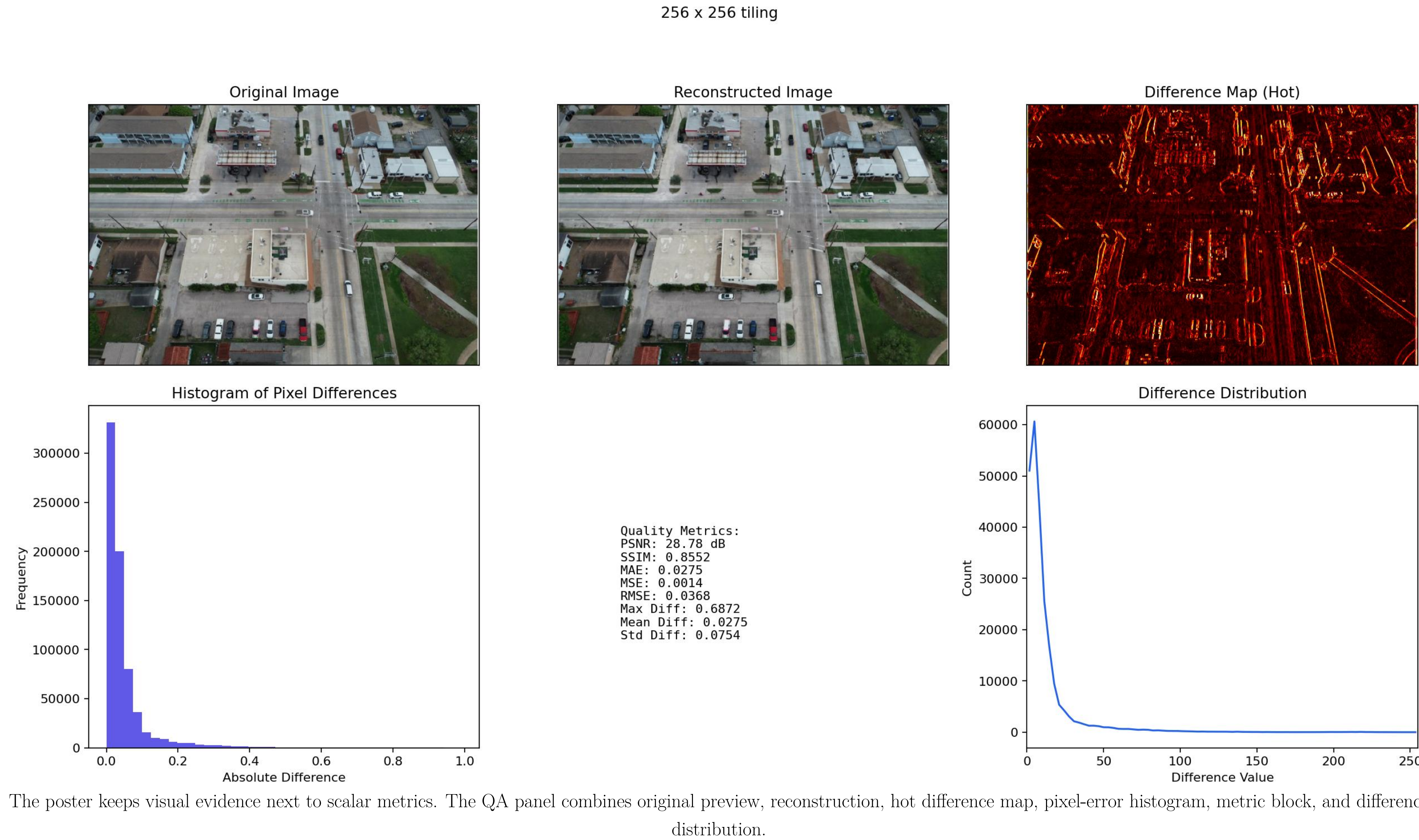
Selected Validation Table

Setup	Sec./img	GPU GB	Ratio	PSNR	SSIM
No tiling	144.35	52.01	$70.11 \times$	28.96	0.8616
256^2 tile	79.84	1.66	$68.24 \times$	28.78	0.8552
512^2 tile	87.71	3.02	$66.31 \times$	28.86	0.8583

Why This Belongs at SC

The contribution is an HPC operating-point study: the same CDC model can be impractical or usable depending on GPU memory pressure, diffusion runtime, and image tiling. The poster reports model quality together with system cost, so reviewers can see the path from algorithm to deployable disaster-imagery workflow.

Visual QA Evidence



Interpretation

- Native full-image CDC is feasible but memory-heavy.**
Baseline decoding takes about 144 seconds per image and peaks near 52 GB.
 - Resolution reduction is fast but quality-limited.**
2K inputs run quickly, but SSIM drops to about 0.689.
 - Tiling is the practical control.**
 256×256 tiling gives the strongest measured runtime and memory improvement with small scalar quality loss.
- Next update.** Add a larger validation run, more visual panels, and downstream disaster-analysis checks.

Final-Version Updates

Before final upload: add a larger selected run, more representative visual QA panels, downstream disaster-analysis checks, and a QR code linking to code, result tables, and a short video explanation.